

Vendor Module – Product Listing Table Structure & Database Tables Overview

School Uniform Marketplace Platform – Based on Technical Specification v1.0 (June 2026)

1. Vendor Module – Table Structure

The Vendor Module stores vendor registration, KYC, GST/PAN, bank settlement, and commission details. This table is referenced by the Product, Order, and Settlement tables via vendor_id.

Field Name	Data Type	Description	Validation / Notes
vendor_id	UUID	Unique vendor identifier	Auto-generated, Primary Key
business_name	VARCHAR(150)	Shop / brand name	Required, unique
owner_name	VARCHAR(100)	Owner full name	Required
email	VARCHAR(150)	Login / contact email	Unique, required
phone	VARCHAR(15)	Mobile number	10 digits, required
address	TEXT	Business address	Required
city	VARCHAR(80)	City of operation	Required
state	VARCHAR(80)	State	Required
pincode	VARCHAR(10)	PIN code	6-digit
gstn	VARCHAR(20)	GST Identification Number	Format: 15-char GSTIN
pan_number	VARCHAR(15)	PAN card number	10-char alphanumeric
bank_account_no	VARCHAR(25)	Bank account for settlement	Encrypted storage
ifsc_code	VARCHAR(15)	Bank IFSC code	11-char format
commission_rate	DECIMAL(5,2)	Platform commission %	Range 0–100
status	ENUM	pending / approved / suspended	Set by Admin
logo_url	VARCHAR(255)	Vendor logo image URL	Optional
created_at	DATETIME	Registration date	Auto-set

2. Vendor Module – Product Listing Table Structure

The Product table is created/owned by the Vendor (via the Vendor Portal's "Product Listing Management" feature). Every product is linked back to a vendor_id and goes through an admin/school approval workflow before becoming visible to parents.

Field Name	Data Type	Description	Validation / Notes
product_id	UUID	Unique product identifier	Auto-generated, Primary Key
vendor_id	FK → vendor_id	Vendor who listed the product	Required (Foreign Key)
product_name	VARCHAR(200)	Product title (e.g. School Shirt Boys)	Required
category	ENUM / FK	shirt / pant / skirt / tie / shoes etc.	Required
gender_type	ENUM	boys / girls / unisex	Required
sizes_available	JSON	Array: [S, M, L, XL, 28, 30...]	At least 1 size required
price	DECIMAL(10,2)	MRP per unit	Positive number
discount_price	DECIMAL(10,2)	Discounted selling price	Less than price, optional

Field Name	Data Type	Description	Validation / Notes
gst_rate	DECIMAL(5,2)	GST percentage (e.g. 5, 12, 18)	Valid GST slab
stock_qty	INT	Total available stock	Non-negative
images	JSON	Array of image URLs	Min 1 image required
description	TEXT	Product description	Optional
fabric_type	VARCHAR(80)	Cotton / Polyester / Blend etc.	Optional
approval_status	ENUM	pending / approved / rejected	Admin / School approval required
is_active	BOOLEAN	Listed / unlisted by vendor	Default: true
created_at	DATETIME	Listing date	Auto-set

3. Vendor Product Listing Flow (Step-by-Step)

Step 1 – Vendor Login

Vendor logs into the Vendor Portal using credentials issued by Admin during onboarding.

Step 2 – Add Product

Vendor fills product details: name, category, gender_type, sizes_available, price, discount_price, gst_rate, stock_qty, images, description, fabric_type. Record inserted into the Product table with vendor_id = current vendor and approval_status = 'pending'.

Step 3 – Admin Review

Admin reviews the pending product in the Admin Panel → Product Approval section. Admin can approve or reject.

Step 4 – Approval Update

On approval, product.approval_status changes to 'approved'. Admin + School are notified by email (per Notification System). Rejected products return to vendor with feedback (implicit via status).

Step 5 – School Mapping

School logs into School Portal and maps the approved product to specific class_name(s) and gender_type via the School–Product Mapping table, setting approved_by_school = true.

Step 6 – Visibility to Parents

Once mapped and approved_by_school = true, the product becomes eligible for the Product Visibility Engine. It appears to parents only if product.is_active = true, vendor.status = 'approved', and product.stock_qty > 0.

Step 7 – Inventory & Order Management

Vendor manages stock_qty per size via Inventory Control. As parent orders come in, vendor processes orders (confirmed → packed → shipped) and updates tracking_number in the Order table.

Step 8 – Sales & Commission Tracking

Vendor views Sales Reports and Commission View — commission_amount is computed per order based on vendor.commission_rate, and net payout is shown ahead of settlement.

Summary Flow: Vendor Login → Add Product (pending) → Admin Approval → School Maps Product to Class/Gender → School Approval → Product Becomes Visible to Matching Parents → Parent Orders → Vendor Processes & Ships → Sales Report & Commission Tracked → Admin Settlement Payout.

4. Total Database Tables Used in This Project

Based on Section 4 ("Database Field Specifications") of the Technical Specification document, the platform is built on the following 8 core tables. Additional supporting tables (notifications, settlements log) are implied by the feature set but not separately detailed in the spec.

#	Table Name	Purpose	Key Relationships
1	Admin	Stores platform admin/sub-admin accounts and roles	Referenced by School.onboarded_by
2	Vendor	Vendor registration, KYC, GST, bank & commission details	Referenced by Product, Order
3	School	School registration, address, classes_offered, status	onboarded_by → Admin; referenced by Student, Mapping
4	Parent	Guardian login accounts (email/phone OTP)	Referenced by Student, Order
5	Student Profile	Child profile linked to a parent + school + class + grade	parent_id → Parent; school_id → School; referenced by C
6	Product	Vendor-listed uniform products (the core catalog)	vendor_id → Vendor; referenced by School–Product Map
7	School–Product Mapping	Core visibility table mapping approved products to class/grade	School_id → School; product_id → Product
8	Order	Order/transaction records with items, payment, GST status	student_id, school_id, vendor_id → respective

Total core tables in this project: 8 (Admin, Vendor, School, Parent, Student Profile, Product, School–Product Mapping, Order). These 8 tables together form the complete relational database schema for the multi-role School Uniform Marketplace platform as described in the specification.

Source: "School Uniform Marketplace – Platform Design & Technical Specification" (Version 1.0, June 2026) and "Kwality Uniform" Business Proposal (Digiware Technology Services, June 2026).